

Hoang Le

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Lansing, MI

TECHNICAL SKILLS

Specialties: Machine Learning, Computer Vision, Vision-Language Models, Large Language Models, Deep Learning.

Machine Learning: Stable Diffusion, Transformer, PyTorch, Colab, Scikit-learn, Pandas, HuggingFace.

Computer Vision: CNN, ViT, Vision Language Model, 3D Object Detection, CLIP, 3D Reconstruction, Semantic Segmentation

Languages: Python, C++, Java.

Tools/Framework: Docker, Github, Linux, OpenCV, FastAPI, HTML/CSS, Linux, React, PostgreSQL, AWS.

EDUCATION

Michigan State University | B.S. in Computer Science | GPA: 3.89 East Lansing, MI Expected 05/2027

Courseworks: Deep Learning (A), Computer Vision (A), Machine Learning (A), Probability and Statistics (A), Linear Algebra (A)

Awards: Dean List, Michigan State University, 2023 - 2026

Research: Le, H., et al. (2026). SiPhy: Single-view physical property reasoning. ECCV 2026

Research: Le, H., et al. (2024). DAM3D: Detect anything in 3D. Poster presentation. MidSURE 2024

Activities: AI Club, Claude Builder Club, Hackathons, East Central North America Regional ICPC

EXPERIENCES

RCL Lab - MBZUAI

Abu Dhabi, UAE

06/2026 - Now

Research Assistant | *Deep Learning, Robotics, 3D Perception, Mujoco*

Worked with Dr. Xingxing Zuo to build a pipeline that retargets human hand-object grasps from video to robotic dexterous hands, enabling robots to imitate human actions from video, removing the needs for expensive teleoperation data collection

- Built a video-to-robot-grasp pipeline combining 3D perception (SAM3, CoTracker, MediaPipe, SAM3D) with contact-conditioned grasp optimization (CEDex-Grasp), cuRobo motion planning, and MuJoCo physics validation, increasing physically-held grasps by **+30%** (0/256 to 75/256) over a contact-agnostic diffusion baseline.
- Debugged and fixed physics-simulation fidelity issues (floor calibration, object mass, mesh collision) causing false-positive "held" grasps, increasing finger-object contact rate by **5-10x** and enabling reliable grasp evaluation across **1,000+** candidates.

Domino Lab - Michigan State University

East Lansing, MI

05/2025 - Now

Research Assistant | *Deep Learning, CLIP, Vision Language Model, 3D Reconstruction, Pytorch*

Worked with Dr. Zijun Cui to lead one of the first single-view models development to **predict physical properties from images**, enabling robotic grasp planning by estimating grasp location and force from inferred material properties.

- Implemented and finetuned **Vision-Language Model** (LLaVA-based) using **Pytorch** for material segmentation, achieving results better than NeRF2Physics (CVPR 2024), PUGS (ICRA 2025), comparable to ChatGPT-4v (IoU: ours 25, GPT4: 23)
- Achieved **state-of-the-art results** on 3 synthetic datasets and 2 Robotics hand-object datasets, lowering mass prediction error by up to **45%**, reducing density MAE by **35.5%**, and lowering Young's modulus error by **23.5%** compared to prior methods.
- First-author paper **accepted** to main track of **ECCV 2026** ([code](#))

CV Lab - Michigan State University

East Lansing, MI

01/2024 - 01/2025

Research Assistant | *Deep Learning, 3D Object Detection, Multi-modal, Zero-shot Learning, Inference pipeline*

Worked with Dr. Abhinav Kumar and Dr. Xiaoming Liu to improve **3D object detection models** for autonomous driving.

- Engineered **unified inference pipeline** combining SAM, metric depth, FoundationPose on RTX 3090, sustaining **18 FPS**, **+44% improvement** compared to Mask RCNN 10 FPS.
- Setup a baseline for **monocular 3d object detection** tasks with **+3%** AP3D higher than SoTA model on ScanNet dataset.
- Submitted and presented a poster at MidSURE 2024, with attendance of over 350 undergrad students from 70 institutions.

PROJECTS

Crack Length Prediction

East Lansing, MI

10/2025 - 12/2025

Technologies: *Computer Vision, Deep Learning, 3D Object Detection, Multi-modal, Zero-shot Learning, Segmentation models*

- **Awarded best project - CSE803: Graduate computer vision course**
- Implemented ViT-based and U-Net crack **segmentation models** from scratch in **Pytorch**, with a segmentation-aware regression head for metric crack length estimation.
- Outperformed **large VLMs** (LLaVA-1.5, Qwen-VL, InstructBLIP, DeepSeek-VL2) on crack length prediction, achieving 20.29 cm RMSE / 13.74 cm MAE, improving over the best VLM by **+36%** RMSE and **+46%** MAE.

NeuralNetBuilder

East Lansing, MI

02/2025 - Present

Technologies: *Machine Learning, React, FastAPI, PyTorch, AWS ECS, Docker, PostgreSQL*

- Building a **cloud-based drag-and-drop ML platform** that enables **visual model composition** and **one-click remote training**, targeting < 5 min time-to-first-run through automated graph validation and job orchestration.
- Implementing a **containerized training pipeline** with **autoscaling** workers and experiment tracking to reduce queue latency and improve training reliability.

AWARDS

- Engineering Summer Undergraduate Research Fellowships, Michigan State University, 2024
- Second Prize, Vietnamese National Mathematics Olympiad, 2022